6a. turunan first principle

diket : $\pi(x) = -(a+1)x^3 + (b+5)x^2 - 100x - (k^2, 10)$

subs : $\pi(x) = -3x^3 + 99x^2 + 100x - 810$

definisi : $\pi'(x) = \lim_{h \to 0} \frac{\pi(x+h) - \pi(x)}{h}$

expansi : $(x+h)^3 = x^3 + 3x^2 h + 3xh^2 + h^3$

$(x+h)^2 = x^2 + 2xh + h^2$

hasil $\pi'(x) = -9x^2 + 198x + 100$

6b. turunan lanjut : $\pi''(x) = -18x + 198$

$\pi'''(x) = -18$

6c. titik kritis :

bagi -1 : $-9x^2 + 198x + 100 = 0$

$9x - 198x - 100 = 0$

rumus kuadrat : $x = \frac{198 \pm \sqrt{198^2 + 3600}}{18} = \frac{198 \pm \sqrt{42804}}{18}$

$= \frac{198 \pm 206.88}{18}$ hasil : $x_1 = 22.49$

$x_2 = -0,49$

6d. gradient descent

rumus : $x_{k+1} = x_k + a\pi'(x_k)$

$x_0 = 1$

iterasi singkat

$k = 0$ $\pi(1) = 289$

$x_0 = 1$ $x_1 = 1 + 0.007(289) = 3.023$

$k = 1$ $\pi(3.023) = 613.3$ $x_2 = 7.316$

$k = 2$ $\pi(7.316) = 1059.6$ $x_3 = 14.733$

$k = 3$ $\pi(14.733) = 1142.2$ $x_4 = 22.728$

Dipindai dengan CamScanner

Date. /

6g. hasil bisnis

$x^* = 2249$

harga : Rp. 22.490

profit maximum = π (22.49) = 25.000 (±)

biaya = 4 · 10 = 90 rb

margin = 22.49 − 9 = 13.49 = Rp 13.490 / unit

7a data & pola